



What is Machine Learning?

Applied Machine Learning

Foundation of Machine Learning

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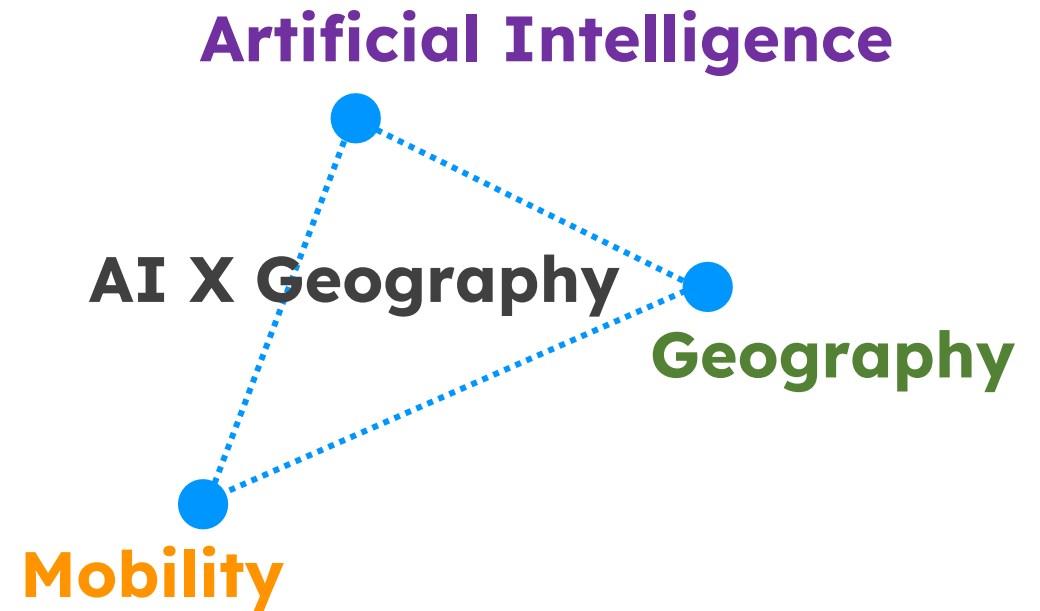
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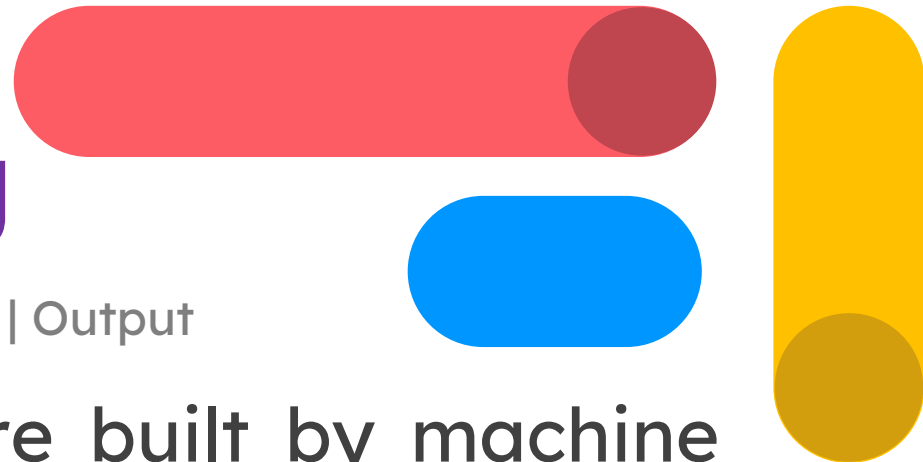
Outline

#GeoAI #Mobility #Geography #STDS

- 01 Why Machine Learning?
- 02 Learning from Data
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- 04 Simple Model
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Why Machine Learning



Clear Use Case | Predictive ML & Data | Prediction Power | Output

Nowadays, several applications were built by machine learning, such as recommendation system (music or video), weather forecasting, mobility prediction, and medical image analysis.

Due to the complexity of data nature, it is hard to identify all simple linear or nonlinear factors/relationship with our target question by knowledge-driven feature engineering.

Machine learning provides an alternative approach to let the machine figure out a way to reveal the mathematical relationship between them.

Why Machine Learning

Clear Use Case | Predictive ML & Data | Prediction Power | Output

From a conventional approach (programming design), ...

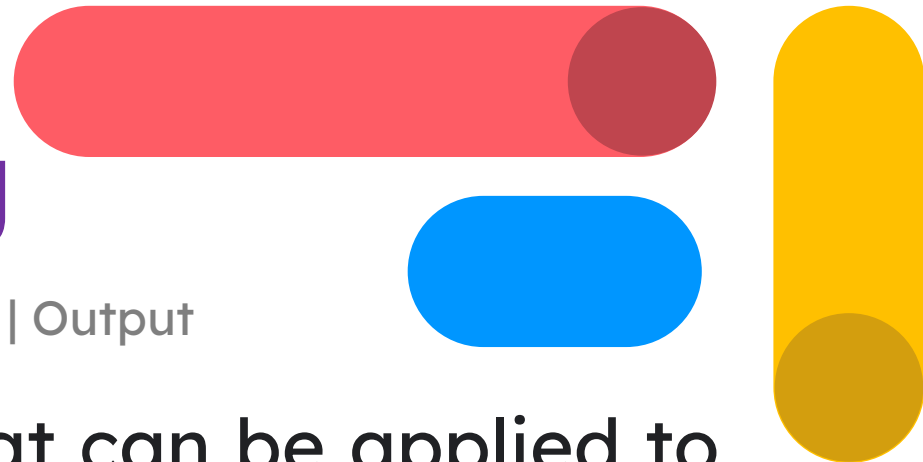
Data + Rules = Answer

Once you know the mathematical formula/ equation, and you may compute your answer without machine learning, such as ...

$$T_F = \frac{9}{5} \times T_C + 32$$
$$T_K = T_C + 273$$

If we know the exact rule, we usually do not need machine learning.

Why Machine Learning



Clear Use Case | Predictive ML & Data | Prediction Power | Output

Some view ML as a universal tool that can be applied to all problems. In reality, ML is a specialized tool suitable only for particular problems.

You don't want to implement a complex ML solution when a simpler non-ML solution will work.

Simple way is the best !

Why Machine Learning

Clear Use Case | **Predictive ML & Data** | Prediction Power | Output

ML systems can be divided into two broad categories: predictive ML and generative AI.

	Input	Output	Training technique
Predictive ML	Text Image Audio Video Numerical	Makes a prediction, for example, classifying an email as spam or not spam, guessing tomorrow's rainfall, or predicting the price of a stock. The output can typically be verified against reality.	Typically uses lots of data to train a supervised, unsupervised, or reinforcement learning model to perform a specific task.
Generative AI	Text Image Audio Video Numerical	Generates output based on the user's intent, for example, summarizing an article, or producing an audio clip or short video.	Typically uses lots of unlabeled data to train a large language model or image generator to fill in missing data. The model can then be used for tasks that can be framed as fill-in-the-blank tasks, or it can be fine-tuned by training it on labeled data for some specific task, like classification.

Why Machine Learning

Clear Use Case | **Predictive ML & Data** | Prediction Power | Output

To confirm that ML is the right approach, first verify that your current non-ML solution is optimized.

The non-ML solution is the benchmark you'll use to determine whether ML is a good use case for your problem.

Quality. How much better do you think an ML solution can be?

Cost and maintenance. How expensive is the ML solution in both the short- and long-term?

Why Machine Learning

Clear Use Case | **Predictive ML & Data** | Prediction Power | Output

Data is the driving force of predictive ML. To make good predictions, you need data that contains features with predictive power.

Abundant	The more relevant and useful examples in your dataset, the better your model will be.
Consistent and reliable	Having data that's consistently and reliably collected will produce a better model.
Trusted	Understand where your data will come from.
Available	Make sure all inputs are available at prediction time in the correct format.

Why Machine Learning

Clear Use Case | **Predictive ML & Data** | Prediction Power | Output

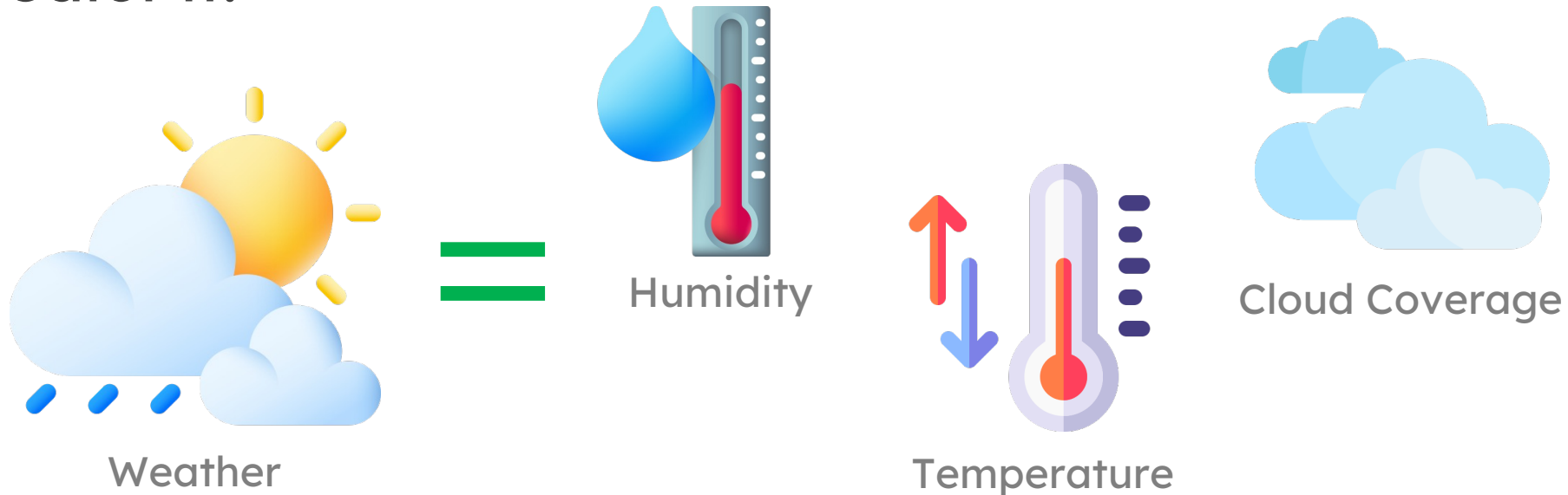
In addition to the volume and sources of the data, correctness and representative are very important to your model.

Correct	In large datasets, it's inevitable that some labels will have incorrect values, but if more than a small percentage of labels are incorrect, the model will produce poor predictions.
Representative	The datasets should be as representative of the real world as possible. In other words, the datasets should accurately reflect the events, user behaviors, and/or the phenomena of the real world being modeled.

Why Machine Learning

Clear Use Case | Predictive ML & Data | **Prediction Power** | Output

For a model to make good predictions, the features in your dataset should have predictive power. The more correlated a feature is with a label, the more likely it is to predict it.



Why Machine Learning

Clear Use Case | Predictive ML & Data | **Prediction Power** | Output

There are two approaches to reveal the relationship between factors and target question.

- **Knowledge-driven**

Well-known relationship

- **Data-driven**

Unknown relationship but can reveal or capture by correlation analysis, e.g., Pearson correlation, Non-negative Matrix Factorization, ...

Why Machine Learning

Clear Use Case | Predictive ML & Data | Prediction Power | **Output**

Determine your model output, ...

Your choice of model type depends upon the specific context and constraints of your problem. The model's output should accomplish the task defined in the ideal outcome. Thus, the first question to answer is "What type of output do I need to solve my problem?"

Why Machine Learning

Clear Use Case | Predictive ML & Data | Prediction Power | **Output**

For a **predictive ML**, ...

	ML system	Example output
Classification	Binary	Classify an email as spam or not spam.
	Multiclass single-label	Classify an animal in an image.
	Multiclass multi-label	Classify all the animals in an image.
Numerical	Unidimensional regression	Predict the number of views a video will get.
	Multidimensional regression	Predict blood pressure, heart rate, and cholesterol levels for an individual.

Why Machine Learning

Clear Use Case | Predictive ML & Data | Prediction Power | **Output**

For a **generative** ML, ...

Model type	Example output
Text	Summarize an article. Reply to customer reviews. Translate documents from English to Mandarin.
Image	Apply visual effects to photos. Generate product design variations.
Audio	Generate dialogue in a specific accent. Generate a short musical composition in a specific genre, like jazz.
Video	Generate realistic-looking videos. Analyze video footage and apply visual effects.
Multimodal	Produce multiple types of output, like a video with text captions.

Learning from Data



Preface | Pattern & Data | Examples

- Before starting the details of how machine identify, knowing, or even understanding the object or our target, we have to think even more backward.
- How do you learn?
- Or more specifically, how do you recognize an object?
- ...
- Basically, all learning algorithms were designed by our thinking rationale ...

Learning from Data

Preface | **Pattern & Data** | Examples

How do you recognize the following animals?



Learning from Data

Preface | **Pattern & Data** | Examples

We learn all items and knowledge from **THE DATA**.

That is the reason why own data is the upmost thing for both laboratories and companies, such as OpenAI and Anthropic.

But what is the power of learning data in the real world?

If you want to travel to London next week, you have to book a hotel or apartment for accommodation.

During the process of choosing target accommodation place, you might refer to the ratings of each hotel or apartment.

Learning from Data

Preface | Pattern & Data | **Examples**

Here is one example from Booking.com

The screenshot shows the Booking.com interface. At the top, there are navigation links for '住宿' (Accommodation), '航班' (Flights), '機票 + 飯店' (Flights + Hotels), '租車' (Car Rental), and '景點 / 活動' (Attractions / Activities). Below these is a search bar with '輸入目的地 倫敦' (Enter destination London) and a date selector. The main content area shows the listing for 'Charming flat with balcony in Pimlico' with a location pin icon and the address '38 Alderney Street, Flat 5, London, SW1V 4EU, 西敏市, 倫敦, SW1V 4EU, 英國 - 地'.

Charming flat with balcony in Pimlico的住客評語

8.5 非常好
65 則評語

我們力求提供 100% 真實評語 ⓘ

各項目評分：



篩選條件

旅客	評分	語言	住宿期間
全部 (65) ▼	全部 (65) ▼	全部 (65) ▼	全部 (65) ▼

Learning from Data

Preface | Pattern & Data | **Examples**

If you forgot to prepare a travel adapter, ...

綜合服務 ▾ 最近看過 購物車 我的訂單 顧客中心 下載 APP 註冊 | 登入

PChome 24h 99 全站 ▾ 除濕機

iphone17 ps5pro 洗衣機 wifi分享器 掃拖機器人 沖牙機 漱口水 洗碗錠 蛋黃酥 vivobook 電視 耳機 護腕 監視器 螞蟥藥

3C | 周邊 | 筆電 | 通訊 | 數位 | 家電 | 日用 | 母嬰 | 食品 | 生活 | 居家 | 休閒 | 保健 | 美妝 | 時尚 | 書店

一日限定 驚爆價

限時瘋搶 一日限定雙品 限時特惠中

Whirlpool W Collection INTELLI-CARE 第六感智能活氧洗衣塔 預購送清淨除濕機

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Learning from Data

Preface | Pattern & Data | **Examples**

If you forgot to prepare a travel adapter, ...

別人也看了

 24h KINYO 多合一雙插座 萬國轉接頭 旅遊... \$599 \$659	 24h KINYO 國際電壓旅行萬國轉接頭旅行組... \$639 \$699	 DTAudio 聆翔【澳規/歐規】多國轉換插頭 澳... \$219 \$399	 24h KINYO 迷你國際電壓旅行萬國轉接頭 雙... \$589 \$649	 24h KINYO 多合一雙插座 GaN氮化鎵 萬... \$669 \$729	 DTAudio 聆翔 多國旅行排插 韓國轉接頭 延... \$513 \$999
 24h KOLIN 歌林 萬國充電器轉接頭 (Type-C... \$549 \$800	 DTAudio 聆翔 雙插座 多國旅行轉接頭 萬用... \$229 \$399	 KINYO 多合一旅行萬國轉接頭 MPP-2345 \$720 \$780	 KINYO MPP-2345-1 多合一旅行萬國轉接頭-黑 \$699 \$799	 24h Travel Blue 藍旅 萬用轉接插頭 多國... \$879 \$999	 DTAudio 聆翔 8合1 多國轉接頭 韓國轉接頭 ... \$469 \$999

How to know?

- 1) Timing
- 2) Keywords
- 3) Regional preference
- 4) Personal preference
- 5) Search history
- 6) Moving trajectory
- 7) ...

Components in ML

Let's formulate the descriptive statement (contextual) into mathematical formula.

Input: x

Input space (all possible inputs x): $\mathcal{X}$

Output space: $\mathcal{Y}$

Unknown target function: $f: \mathcal{X} \rightarrow \mathcal{Y}$

Dataset of Input-output examples: $\mathcal{D}: (x_1, y_1), \dots, (x_N, y_N)$

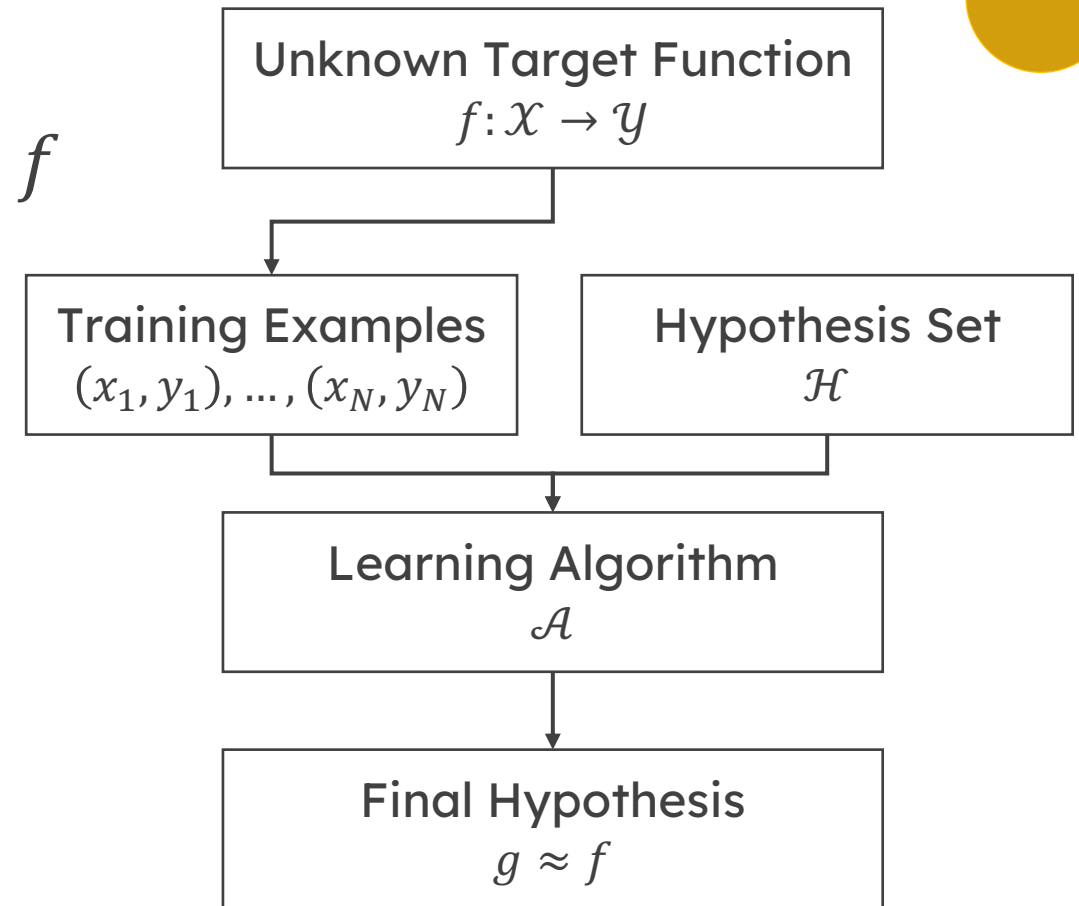
$y_n = f(x_n)$ for $n = 1, \dots, N$

Components in ML

Then, what will we do?

Let $g: \mathcal{X} \rightarrow \mathcal{Y}$ approximates to f

The algorithm choose g from a set of candidate formulas under consideration, which we call the hypothesis set $\mathcal{H}$.



Simple Model

Let $\mathcal{X} = \mathbb{R}^d$ be the input space

$\mathbb{R}^d$ is the d-dimensional Euclidean space

Let $\mathcal{Y} = \{+1, -1\}$ be the output space; the binary output y corresponds to approving or denying credit.

Approve credit if

$$\sum_{i=1}^d w_i x_i > threshold$$

Deny credit if

$$\sum_{i=1}^d w_i x_i < threshold$$

Simple Model

This formula can be written more compactly as

$$h(x) = \text{sign} \left(\left(\sum_{i=1}^d w_i x_i \right) + b \right)$$

The weights are $w_1, \dots, w_d$ and the threshold is determined by the bias term b , credit is approved if $\sum_{i=1}^d w_i x_i > b$.

This model of $\mathcal{H}$ is called the perceptron.

$$\sum_{i=1}^d w_i x_i = w^T x \Rightarrow h(x) = \text{sign}(w^T x)$$

Simple Model

We introduce the perceptron learning algorithm (PLA).

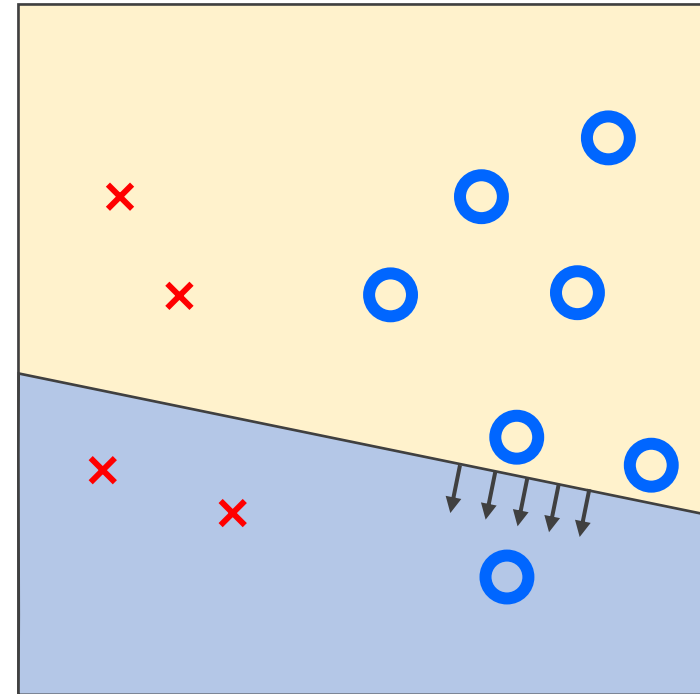
At each iteration $t = 0, 1, 2, \dots$, there is a current value of the weight vector, call it $w(t)$.

The algorithm picks an example from $(x_1, y_1), \dots, (x_N, y_N)$ that is currently misclassified, called it $(x(t), y(t))$, and uses it to update $w(t)$.

Simple Model

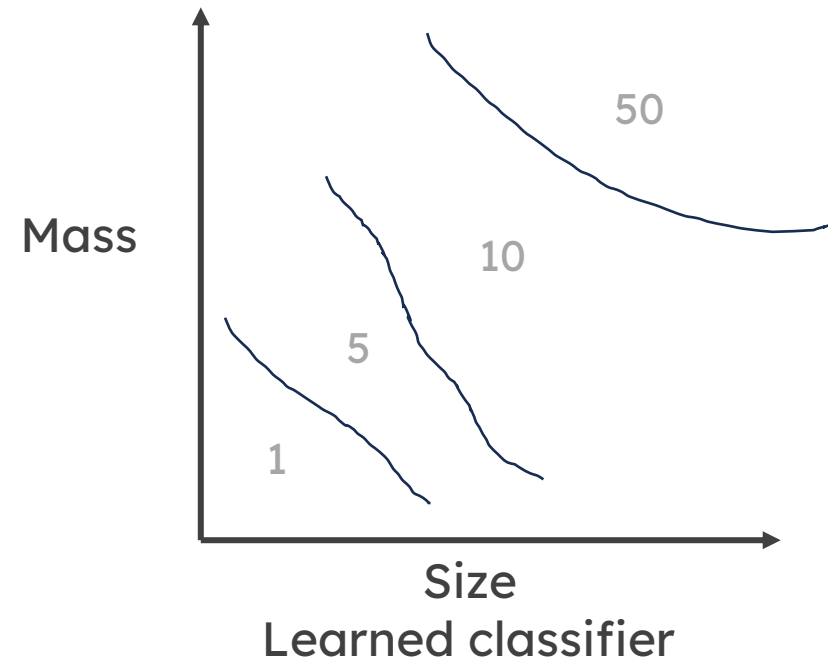
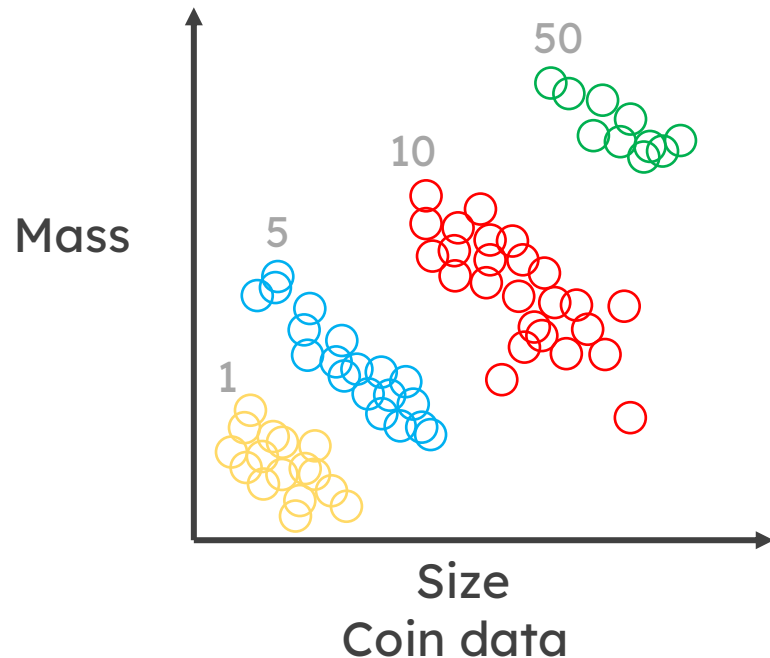
Since the example is misclassified, we have $y(t) \neq \text{sign}(w^T(t)x(t))$. The update rule is

$$w(t+1) = w(t) + y(t)x(t)$$



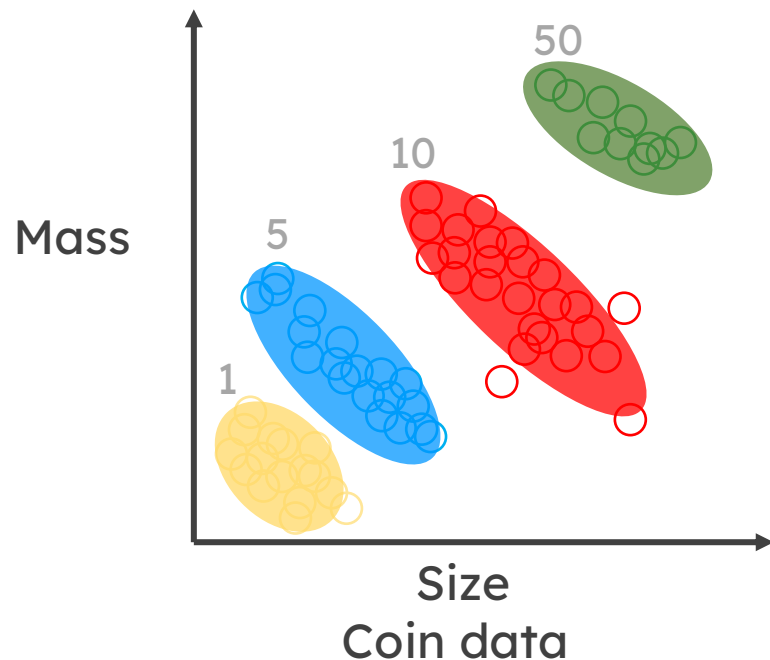
Learning vs Design

After understanding what learning is, now, we introduce what it is not.

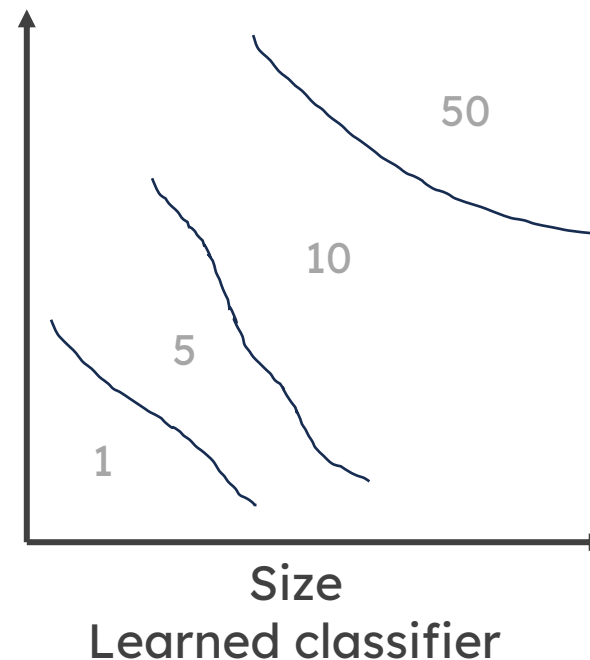


Learning vs Design

The main difference between the learning approach and the design approach is the role that data plays.



The problem is well specified and one can analytically derived f without the need to see any data.



The problem is much less specified, and one needs data to pin down what f is.

Types of Learning

Framework | Supervised | Unsupervised | Self-supervised |
Reinforcement | Semi-supervised

Basically, we can separate all learning algorithms into five types:

- 1) Supervised learning
- 2) Unsupervised learning
- 3) Self-supervised learning
- 4) Reinforcement learning
- 5) Semi-supervised learning

Types of Learning

Framework | **Supervised** | Unsupervised | Self-supervised |
Reinforcement | Semi-supervised

Supervised Learning

Supervised machine learning is a type of machine learning where the model is trained on a labeled dataset (i.e., the target or outcome variable is known).

Supervised learning is commonly used for risk assessment, image recognition, predictive analytics and fraud detection, and comprises several types of algorithms.

Types of Learning

Framework | **Supervised** | Unsupervised | Self-supervised |
Reinforcement | Semi-supervised

Supervised Learning :: Regression



Regression algorithms predict output values by identifying linear relationships between real or continuous values (e.g., temperature, salary).

Regression algorithms include linear regression, random forest and gradient boosting, as well as other subtypes.

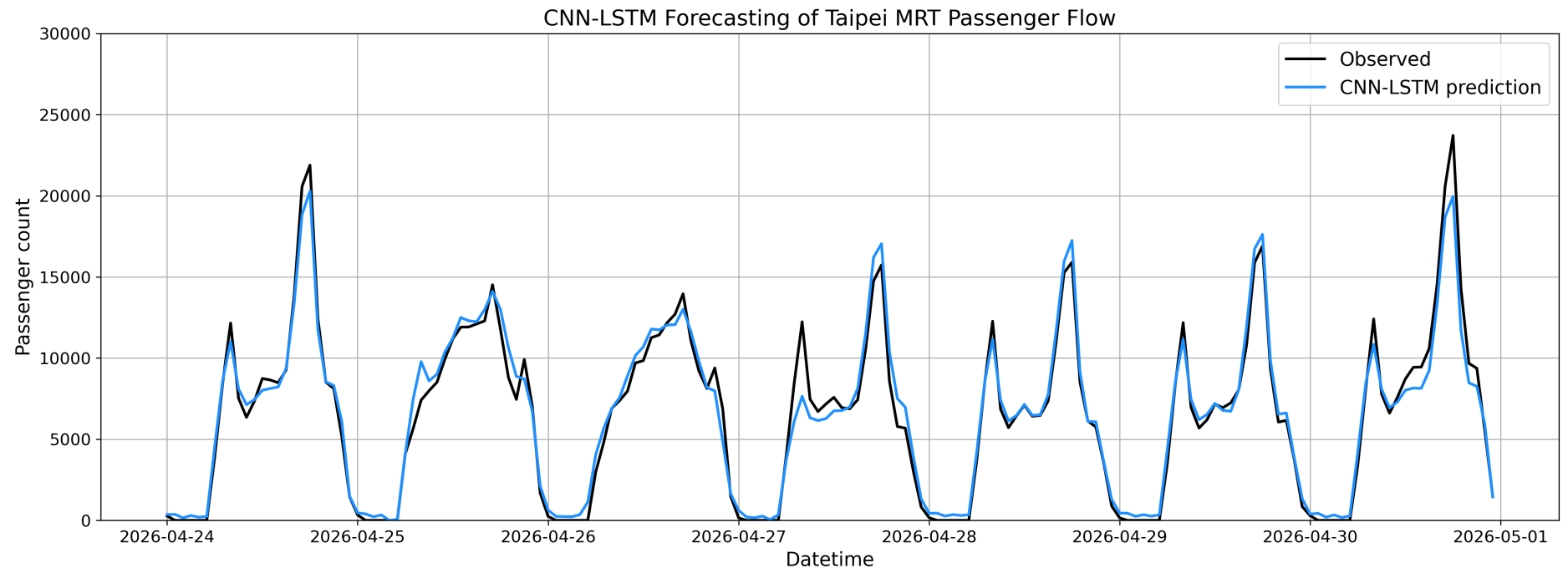
Types of Learning

Framework | **Supervised** | Unsupervised | Self-supervised |
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Supervised Learning :: Regression



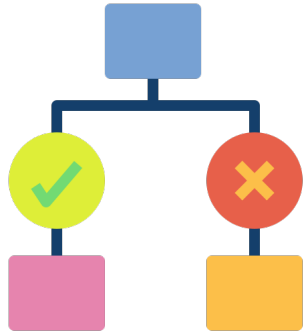
Passenger Flow Prediction



Types of Learning

Framework | **Supervised** | Unsupervised | Self-supervised |
Reinforcement | Semi-supervised

Supervised Learning :: Classification

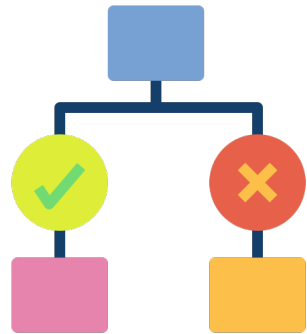


Classification algorithms predict categorical output variables (e.g., “junk” or “not junk”) by labeling pieces of input data. Classification algorithms include logistic regression, k-nearest neighbors and support vector machines (SVMs), among others.

Types of Learning

Framework | **Supervised** | Unsupervised | Self-supervised | Reinforcement | Semi-supervised

Supervised Learning :: Classification



Car Plate Recognition

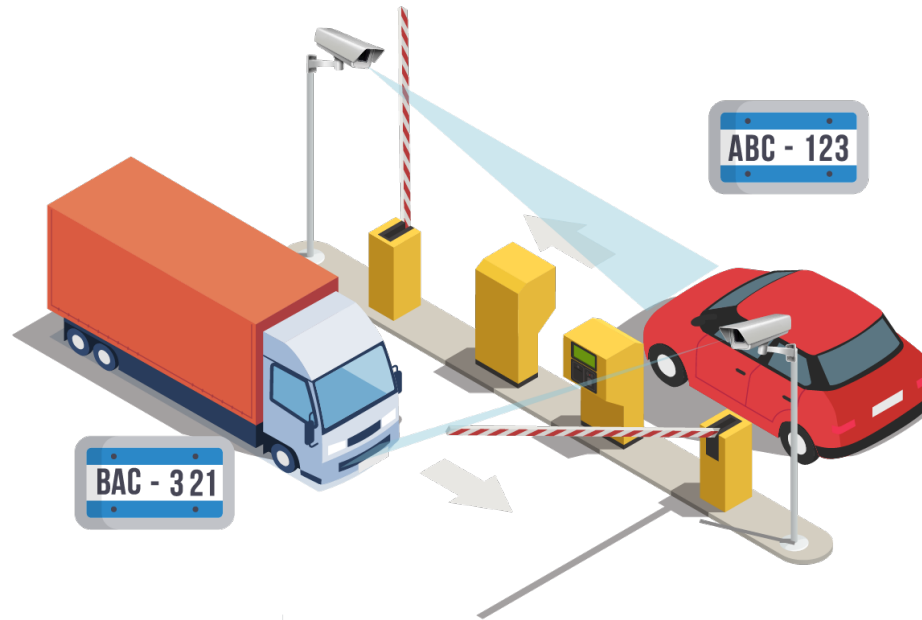


Photo credits: <https://exigasoftware.com.sg/our-products/vehicle-plate-recognition-system-visitor-management-systemlpr-vms/>

Chun-Hsiang Chan (2026)

Face Mask Detection

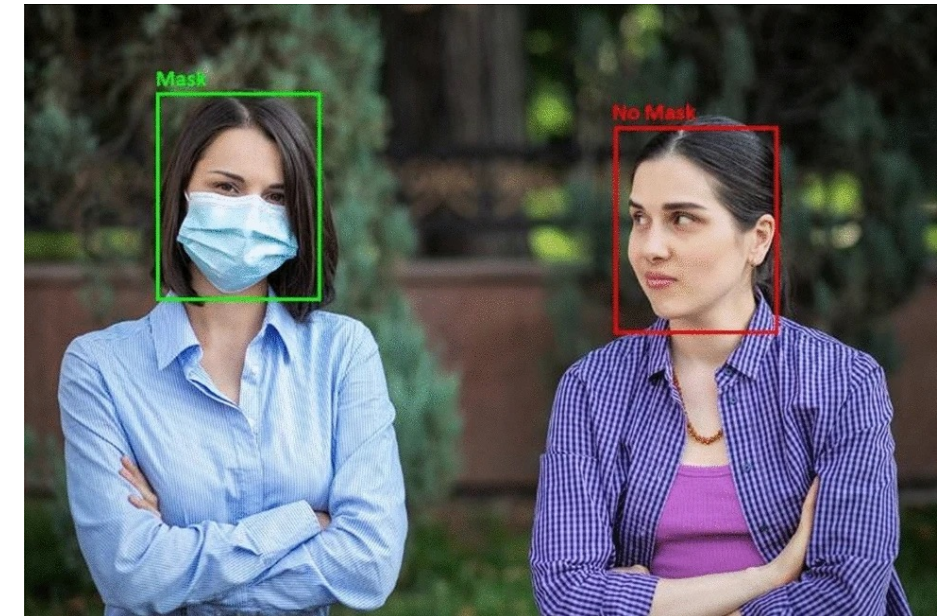


Photo credits: Elhanashi et al. (2023). An integrated and real-time social distancing, mask detection, and facial temperature video measurement system for pandemic monitoring. *Journal of Real-Time Image Processing*, 20(5), 95.

Types of Learning

Framework | Supervised | **Unsupervised** | Self-supervised |
Reinforcement | Semi-supervised

Unsupervised learning

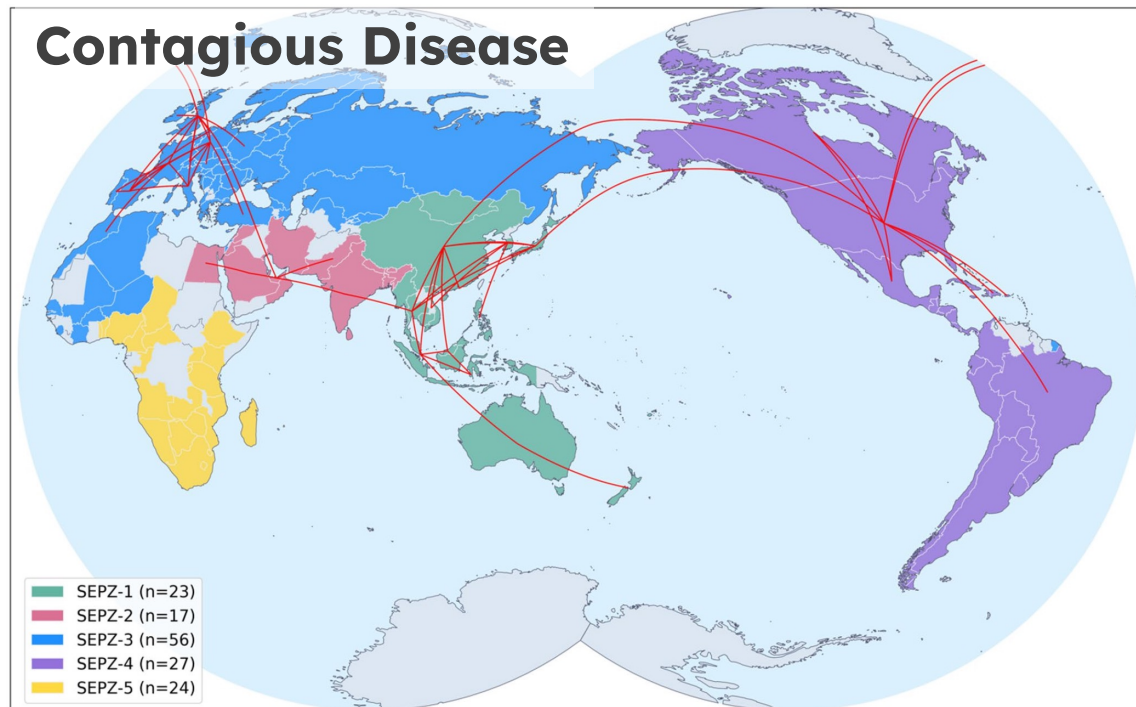
Unsupervised learning algorithms, like Apriori, Gaussian Mixture Models (GMMs) and principal component analysis (PCA), draw inferences from unlabeled datasets, facilitating exploratory data analysis and enabling pattern recognition and predictive modeling.

The most common unsupervised learning method is cluster analysis, which uses clustering algorithms to categorize data points according to value similarity (as in customer segmentation or anomaly detection).

Types of Learning

Framework | Supervised | **Unsupervised** | Self-supervised |
Reinforcement | Semi-supervised

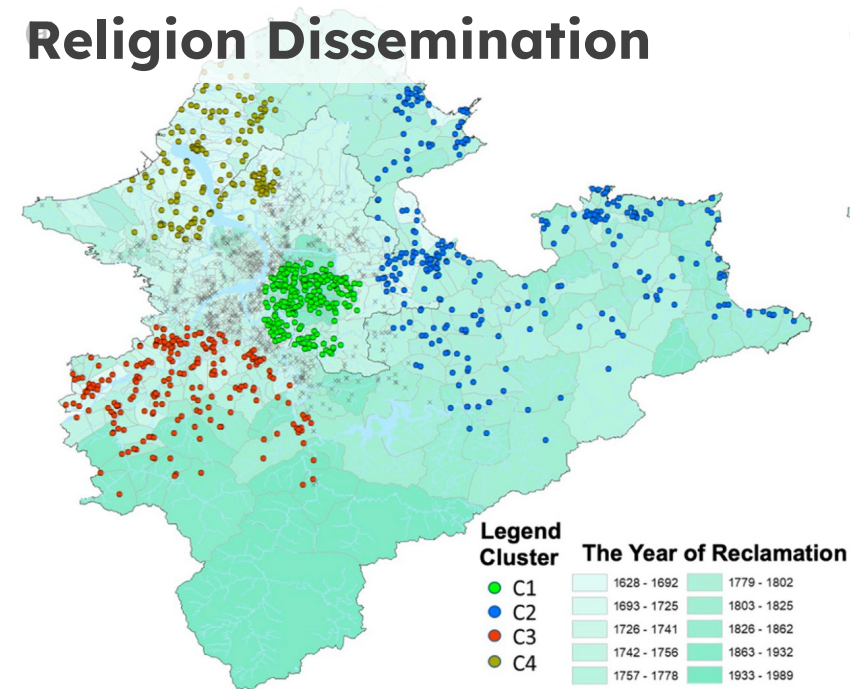
Unsupervised learning :: Clustering



Yang et al. (2026). Understanding the role of international flight networks in disease spread: spatial epidemic prevention zones and hierarchical disease control policies. *International Journal of Health Geographics*.

Fall 2026

Religion Dissemination



Chan et al. (2025). Spatiotemporal topology of religious spread in a multi-religious metropolis: A historical religious profile of Taipei City in Taiwan from 1660 to 2020. *The Geographical Journal*, 191(1), e12609.

Chun-Hsiang Chan (2026)

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Types of Learning

Framework | Supervised | **Unsupervised** | Self-supervised | Reinforcement | Semi-supervised

Unsupervised learning :: Association

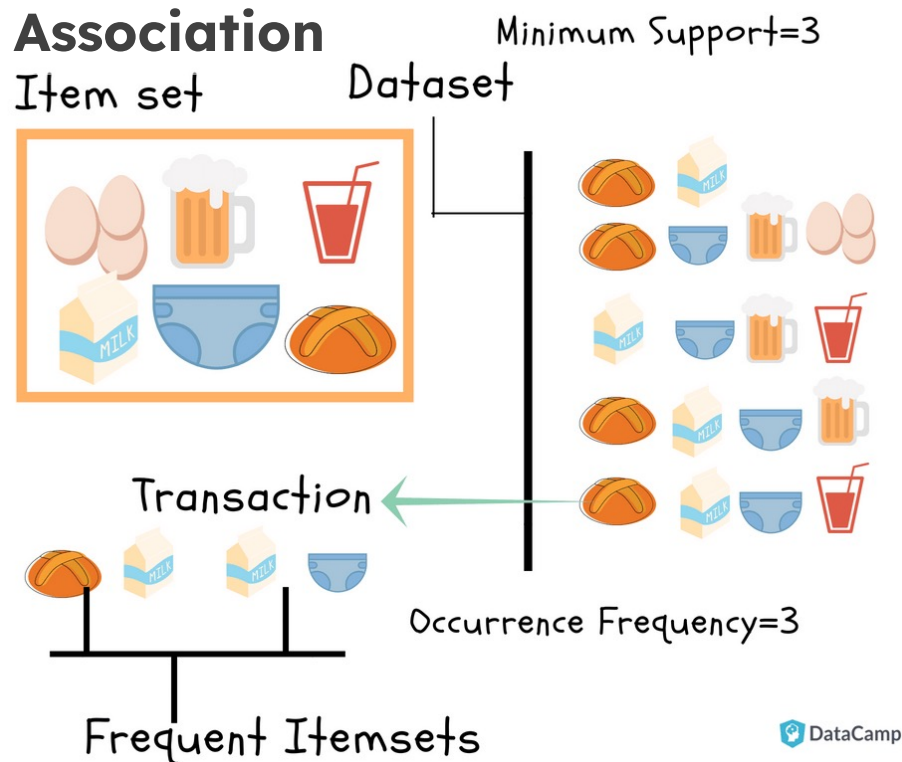
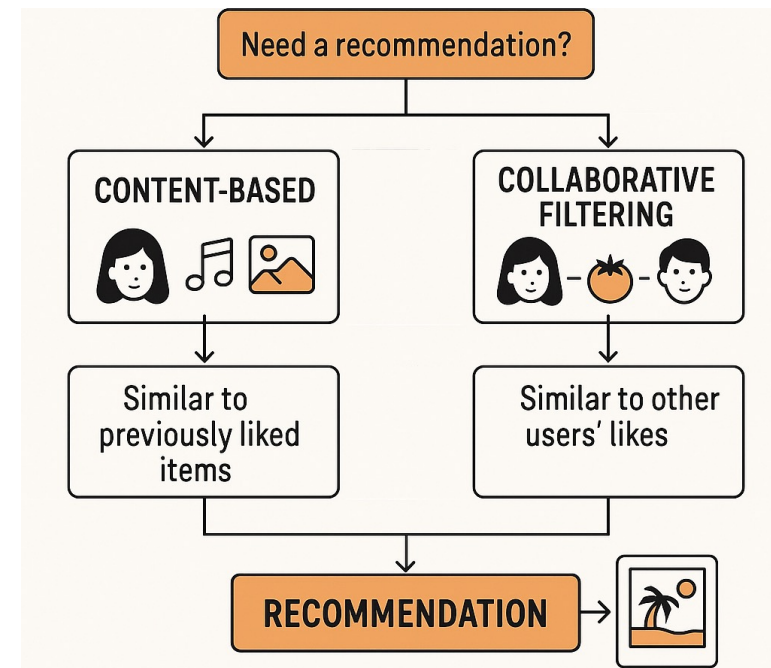


Photo credits: https://en.wikipedia.org/wiki/Affinity_analysis

Recommendation System



Phot credits: <https://generativeai.pub/why-the-internet-feels-so-personal-the-power-of-recommender-systems-de029159a12b>

Types of Learning

Framework | Supervised | Unsupervised | **Self-supervised** |
Reinforcement | Semi-supervised

Self-supervised learning

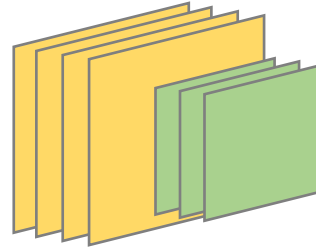
Self-supervised learning (SSL) enables models to train themselves on unlabeled data, instead of requiring massive annotated and/or labeled datasets. SSL algorithms, also called predictive or pretext learning algorithms, learn one part of the input from another part, automatically generating labels and transforming unsupervised problems into supervised ones.

Types of Learning

Framework | Supervised | Unsupervised | **Self-supervised** |
Reinforcement | Semi-supervised

Self-supervised learning

Pretrain

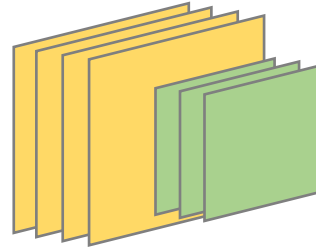


Representation



Pretext
Task

Finetune



Representation



Dog

Types of Learning

Framework | Supervised | Unsupervised | Self-supervised |
Reinforcement | Semi-supervised

Reinforcement learning

Reinforcement learning, also called reinforcement learning from human feedback (RLHF), is a type of dynamic programming that trains algorithms using a system of reward and punishment. To deploy reinforcement learning, an agent takes actions in a specific environment to reach a predetermined goal.

Types of Learning

Framework | Supervised | Unsupervised | Self-supervised |
Reinforcement | Semi-supervised

Reinforcement learning

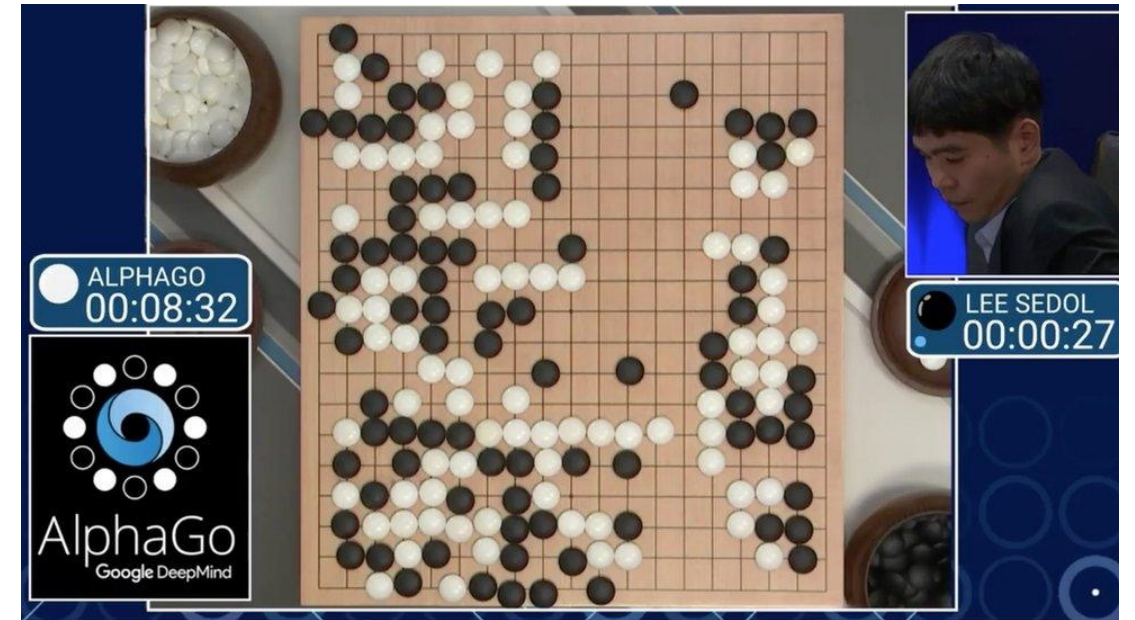
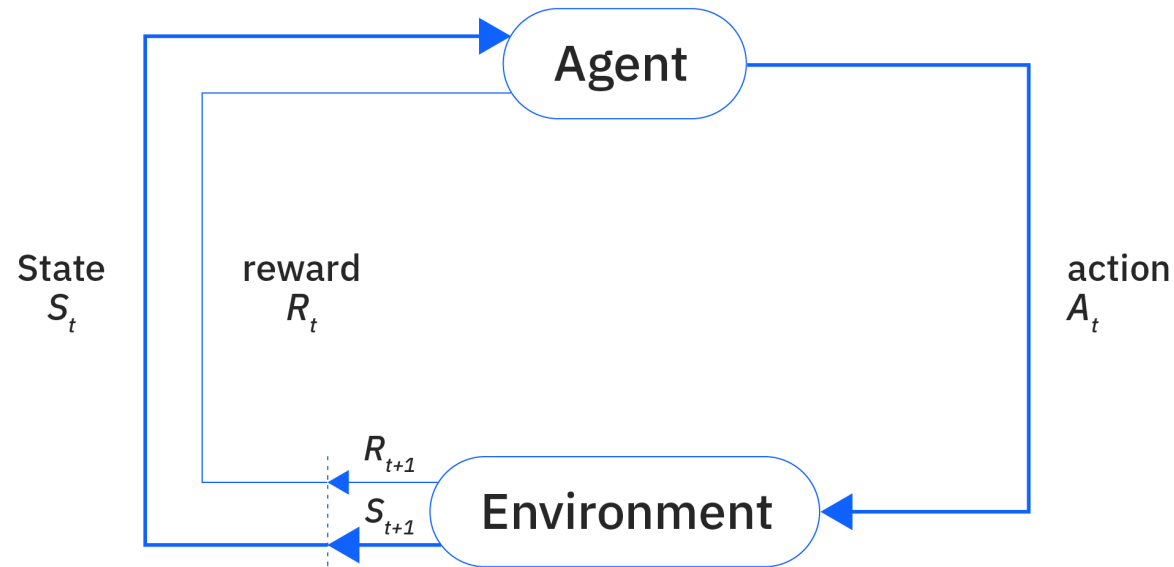


Photo credits: <https://www.bbc.com/news/technology-35785875>

Types of Learning

Framework | Supervised | Unsupervised | Self-supervised |
Reinforcement | **Semi-supervised**

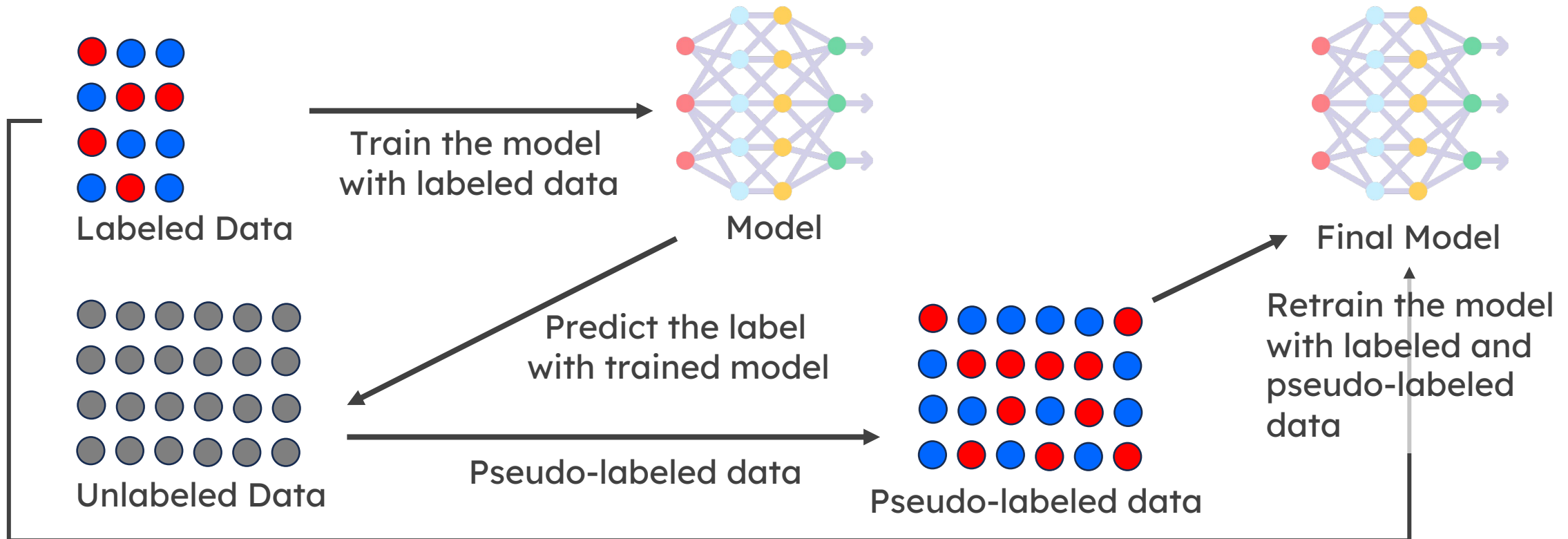
Semi-supervised learning

Semi-supervised learning algorithms are trained on a small labeled dataset and a large unlabeled dataset, with the labeled data guiding the learning process for the larger body of unlabeled data. A semi-supervised learning model might use unsupervised learning to identify data clusters and then use supervised learning to label the clusters.

Types of Learning

Framework | Supervised | Unsupervised | Self-supervised |
Reinforcement | **Semi-supervised**

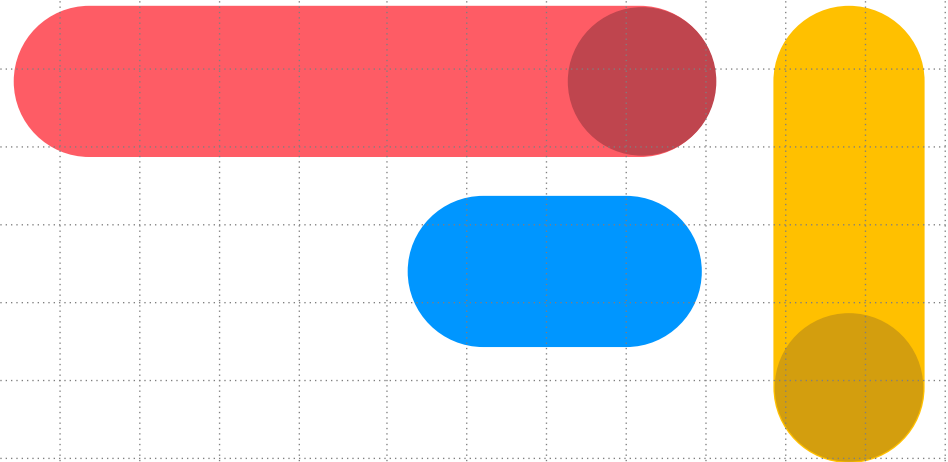
Semi-supervised learning



References

If you want to learn more in details, please see the following references.

- 1) Google Machine Learning :: Foundational courses
- 2) IBM Machine Learning
- 3) Abu-Mostafa, Y. S., Magdon-Ismail, M., & Lin, H. T. (2012). *Learning from data* (Vol. 4, p. 4). New York: AMLBook.



The End

Thank you for your attention!

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